

Lab 10

MySQL

In this lab session, we are going to create a simple database using MySQL.

Our database will keep track of movies that different users have watched. The ER diagram is the following:

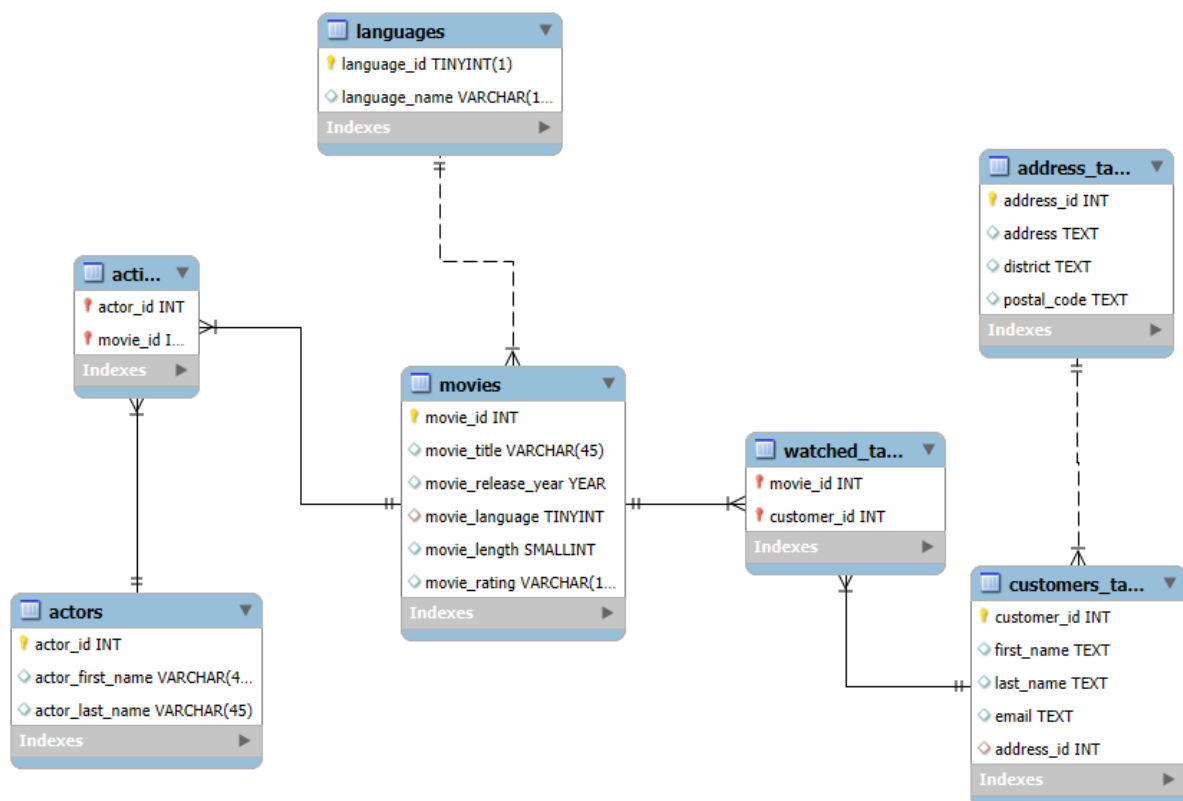


Figure 1 Movies ER Diagram (adaptation from sakila sample DB)

TASK 1 CREATE DATABASE AND TABLES

Start creating manually the tables movies, languages and actors. Then import the data into the created tables.

For the rest of the tables, import the data from the csv files and create a new table for each in the process. After doing this you will need to check and update the data types. Create the FK constraints while updating the tables.

Note 1: Create all tables related to strong entities first, then create the bridge tables. Tables need to be fully defined before creating the FKs (all datatypes and PKs).

Note 2: When creating the Foreign Key constraint, you should use a unique name for the constraint, for example movie_id_acting, movie_id_watched. This does not affect the name of the column/attribute which will still be movie_id in all three tables.

TASK 2 DESCRIBE THE FOLLOWING QUERIES**Query 2.1**

```
SELECT
    movies.movie_title, languages.language_name
FROM
    movies INNER JOIN languages ON
    movies.movie_language=languages.language_id;
```

Query 2.2

```
SELECT movies.movie_title, actors.actor_last_name , actors.actor_first_name
FROM(
    movies inner JOIN acting USING (movie_id)
);
```

Query 2.3

```
SELECT movies.movie_title, actors.actor_last_name
FROM(
    movies INNER JOIN acting USING (movie_id)
    INNER JOIN actors USING (actor_id)
)
WHERE movie_title="ALABAMA DEVIL";
```

TASK 3 QUERIES

Perform the following queries:

1. Get the titles of all the movies stored in our database sorted alphabetically.
2. Get name and address of all customers in our database sorted by last name.
3. Get title and language for the first 20 movies alphabetically.
4. Get title and language of movies released before 2000.
5. Get the average length of movies per year.
6. Get the average length of movies per rating.
7. Get movie titles for movies in German released after 2015.
8. Get movie titles and actors' name and surname for all movies with a G rating.
9. Get movie titles for all movies watched in the district of Texas.